

Answer **all** the questions in the spaces provided.

- 1 (a) Define *velocity*.

Rate of change of displacement

[1]

- (b) The speed v of a sound wave through a gas of pressure P and density ρ is given by the equation

$$v = \sqrt{\frac{kP}{\rho}}$$

where k is a constant that has no units.

An experiment is performed to determine the value of k . The data from the experiment are shown in Fig. 1.1.

quantity	value	uncertainty
v	$3.3 \times 10^2 \text{ m s}^{-1}$	$\pm 3\%$
P	$9.9 \times 10^4 \text{ Pa}$	$\pm 2\%$
ρ	1.29 kg m^{-3}	$\pm 4\%$

Fig. 1.1

- (i) Use data from Fig. 1.1 to calculate k .

$$\frac{(3.3 \times 10^2)^2 \times 1.29}{9.9 \times 10^4} = 1.419$$

$$k = 1.4 \quad [2]$$

- (ii) Use your answer in (b)(i) and data from Fig. 1.1 to determine the value of k , with its absolute uncertainty, to an appropriate number of significant figures.

$$k = \frac{v^2 \times \rho}{P}$$

$$\% \text{ Uncertainty} = (3 \times 2) + 2 + 4 = 12 \%$$

$$\text{Absolute Uncertainty} = 1.419 \times \frac{12}{100} = 0.17028 \approx 0.2$$

$$k = 1.4 \pm 0.2 \quad [3]$$

[Total: 6]